Machine Learning Project 1 — Rush Hour: Predicting Bike-Sharing Demand

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1 Introduction and data

The task is to predict the hourly number of bike rentals (*cnt*) of a bike-sharing system and, in a final step, to classify hours as high demand. For the operator, an under-supplied busy hour is the costly error, so every regression model was assessed not only by R^2 and RMSE but also by the mean shortfall in the busiest hours: hours whose actual demand exceeds the training 90th percentile (452 bikes), and the decile of hours the model itself predicts busiest.

The labelled file contains 10,886 hours on 456 days (days 1–19 of each month, 2011–2012); the hidden test file contains 574 hours, all on day 20. The team seed is the SHA-256 hash of the sorted, underscore-joined student IDs taken modulo 100,000, which gives 36954 (the brief's example value 41,698 is reproduced first). It defines the single 80/20 split into 8,708 training and 2,178 validation hours. All encoders, scalers, centring means, interaction choices and label cutoffs were fitted on the training portion only. Three properties of the data shaped the design: *workingday* equals Monday–Friday minus *holiday* exactly, *season* is an exact function of *mnth* in this file, and the year, month and weekday of *dteday* coincide with *yr*, *mnth* and *weekday*. Each phase consumes an artifact saved by the previous one. Expectations were written before the runs of Phases 2–5; those of Phase 1 and the bonus were reconstructed afterwards and are labelled as such in the notebook.

2 The five-phase chain

2.1 Phase 1: gradient descent

Method. A linear model with 47 main effects was fitted by full-batch gradient descent on $J(w) = \|Xw - y\|^2/(2n)$ with gradient $X^\top(Xw - y)/n$. Hour, weekday, weather situation and month were one-hot encoded with one reference level each, because a numeric hour would force a straight line through a day with two working-day peaks. The weather level 4, absent from the training portion, was merged into level 3. *workingday*, *season*, *atemp* and *instant* were excluded as exact or near copies of other columns. Month was preferred to season (validation R^2 0.6823 against 0.6805). The elapsed-day trend from *dteday* added only 0.0005 to R^2 while raising the condition number of $X^\top X/n$ from 56 to 8,084 (36,407 instead of 512 updates), and was therefore omitted. All features were standardised. Since the cost is quadratic, each update multiplies the error along the steepest direction by $1 - \eta\lambda_{max}$, so gradient descent is stable only for $\eta < 2/\lambda_{max}$; here $\lambda_{max} = 1.9894$ and the limit is 1.0053. Learning rates from $0.005/\lambda_{max}$ to $2.1/\lambda_{max}$ were compared (Figure 1); convergence was declared when $\max|\nabla J| < 10^{-4}$, with a cap of 50,000 updates. The loop uses the equivalent precomputed form $\nabla J = Hw - b$ with $H = X^\top X/n$ and $b = X^\top y/n$, and every fit asserts that both forms agree.

Results. The rate $2.1/\lambda_{max}$ diverged after 84 updates; $0.005/\lambda_{max}$ reached the cap with $\max|\nabla J| = 0.0104$, although its cost curve appeared flat. The chosen rate $1.9/\lambda_{max} = 0.955$ converged in 267 updates, within 0.0074 bikes of the exact least-squares solution. Training R^2 was 0.6970 and validation R^2 0.6823 (RMSE 103.6): the model underfits. Hours above 452 bikes were under-predicted by 176.6 bikes on average, the hours predicted to be busiest by 95.9, and 213 validation hours were predicted below zero. **Passed on:** the weight vector, learning rate, update count, feature list and fitted scaler. Of the training residual variance, 44.4% lay between hour × working-day cells.

2.2 Phase 2: polynomial regression

Method. Interaction groups were selected from Phase 1's training residuals: candidate groupings were ranked by the share of residual variance lying between group means (0.444 for hour × working day), adjusted for the number of groups and checked for sparse groups, and re-ranked after removing the confirmed groups. Degree was interpreted as interaction order, since powers of 0/1 indicators are uninformative. Products were formed from explicit term lists, never multiplying two levels of the same column. Every non-hour factor was centred at its training mean before multiplication, which leaves the representable predictions unchanged and lowered the condition number of the full design from 21,616 to 1,713. Each group was added in turn, every fit warm-started from the Phase 1 weights with new weights at zero (identical starting predictions were asserted) and run with the same $1.9/\lambda_{max}$ rule.

Results. Table 1 lists every step. Hour × working day alone raised validation R^2 by 0.149. Degree 3 reached 0.9225 (RMSE 51.1, 242 features, 3,884 updates). Its train–validation R^2 gap of −0.0047 reflects sampling

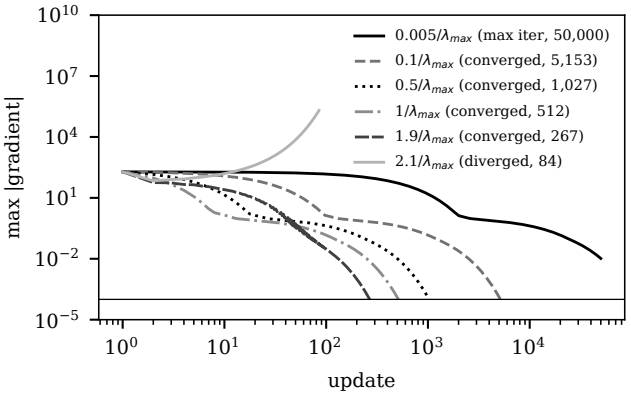


Figure 1: Phase 1 learning-rate comparison: max $|\nabla J|$ per update (log–log); horizontal line: tolerance 10^{-4}.

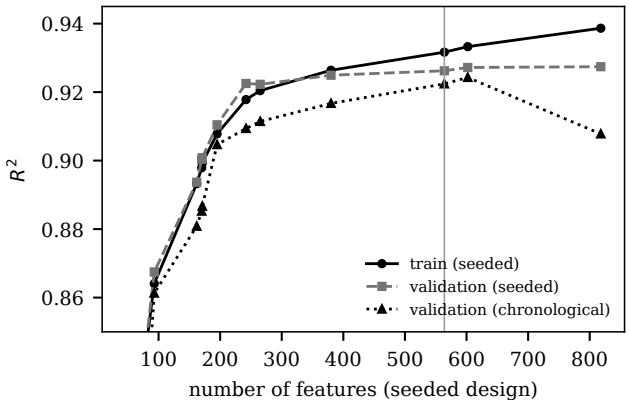


Figure 2: Phase 3 complexity sweep on both splits; vertical line: the 564-feature target.

Table 1: Phase 2: groups added in turn (validation unless marked).

Model	Degree	Features	Train R^2	Val. R^2	Top-predicted-decile shortfall
Phase 1 (main effects)	1	47	0.6970	0.6823	134.9
+ hour × workingday	2	70	0.8316	0.8310	88.0
+ hour × yr	2	93	0.8641	0.8675	77.4
+ hour × season	2	162	0.8933	0.8937	56.5
+ hour-block × weathersit	2	170	0.8979	0.9004	54.1
+ temp × hum	2	171	0.8993	0.9009	53.8
+ hour × workingday × yr	3	195	0.9078	0.9105	50.4
+ hour × workingday × temp	3	242	0.9178	0.9225	43.3

variation, as training and validation RMSE are nearly equal (51.74 and 51.14). The shortfall in the top predicted decile fell from 134.9 to 43.3 bikes. **Passed on:** degree 3 and the 242-feature list with lineage, weights and scaler.

2.3 Phase 3: bias–variance trade-off

Method. The complexity sweep followed Phase 2's order and continued beyond it (hour × holiday, hour × weekday, hour × month, full hour × weather, and the degree-4 group hour × working day × year × season) in order to reach the high-variance regime. Columns that were exact linear combinations of earlier ones were detected by QR factorisation and removed. Learning curves were computed for the degree-3 design and the largest design. The single chronological split held out the last 20% of labelled days (91 days, 2,183 hours, from 2012-08-05) as whole days, with all preprocessing refitted on the earlier days. The target rule was fixed before the run: the converged fit with the fewest features whose seeded validation R^2 lies within 0.002 of the best.

Results. Seeded validation R^2 rose from 0.9225 to a plateau of 0.9274 at 818 features, while the train–validation gap widened to 0.0112 (Figure 2); all fits converged (at most 11,761 updates). The degree-3 learning curve was flat with training and validation RMSE nearly equal, indicating a bias-limited model. The largest design showed a gap of 26.0 bikes at 25% of the data that closed to 4.8, indicating a variance-limited model. On the chronological split degree 3 scored R^2 0.9094 (RMSE 65.5), because it extrapolates into later and busier days. The extensions up to full hour × weather improved the chronological score by 0.0149 (seeded: 0.0047), whereas the degree-4 group reduced it by −0.0165 and increased the top-decile shortfall by 11.2 bikes: the earlier days contain no 2012 autumn hours, so these terms cannot be learned (770 instead of 818 features remain). Combinations observed in training therefore transferred; unobserved ones did not. **Diagnosis:** degree 3 is bias-limited and variance appears only at degree 4. **Passed on:** the 564-feature design ending at hour × month (seeded R^2 0.9262, chronological 0.9224) and the noise candidate temp × hum.

2.4 Phase 4: regularisation

Method. All methods minimised $J(w) = \|Xw - y\|^2/(2n) + \lambda\big[\rho\sum_{j\geq 1}|w_j| + (1-\rho)\sum_{j\geq 1} w_j^2/2\big]$ with an unpenalised intercept: Ridge ($\rho = 0$), Lasso ($\rho = 1$) and Elastic Net ($\rho \in \{0.1, 0.5, 0.9\}$). Each update takes a gradient step on the smooth part, in which the Ridge term adds $\lambda(1-\rho)$ to the diagonal of H, and then soft-thresholds: every penalised weight is moved towards zero by $\eta\lambda\rho$ and set to exactly zero if it would cross it. Ridge used $\eta = 1.9/L$ and the L1-type methods $\eta = 1/L$, with $L = \lambda_{max} + \lambda(1-\rho)$, so that the objective cannot increase. Each method followed a log-spaced λ path from its largest value downwards (for Lasso, from $\lambda_{max} = 70.6$, where all weights vanish), every fit warm-started from the previous one. The brief fixes a single

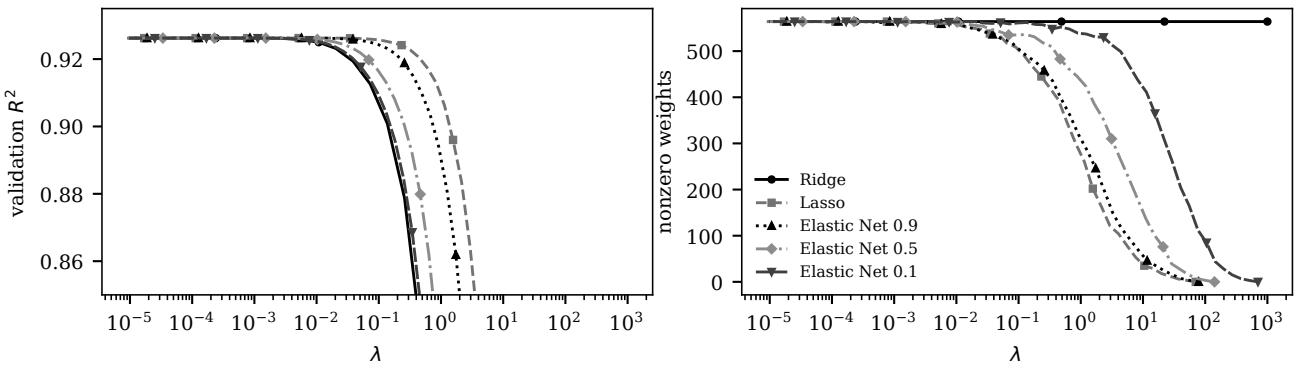


Figure 3: Phase 4 regularisation paths: validation R^2 and number of nonzero weights against λ.

Table 2: Phase 4: selected models (validation; shortfall: mean bikes missing in top-decile hours).

Method	λ	ρ	Nonzero weights	Val. R^2	RMSE	Top-decile shortfall
Unpenalised (Phase 3)	0	–	564	0.9262	49.9	53.8
Ridge	0.000452	0	564	0.9262	49.9	54.0
Lasso	0.169	1	466	0.9248	50.4	56.9
Elastic Net	0.0144	0.1	564	0.9246	50.5	57.6

Table 3: Column verdicts (ΔR^2: change in validation R^2 against the Ridge model; Lasso: surviving terms derived from the column).

Column	Test	ΔR^2	Alone R^2	Lasso	Verdict
`instant`	add back	0.0003	0.094	–	redundant
`dteday`	add back	0.0002	0.094	–	redundant
`season`	removal	0.0000	0.053	56/69	redundant
`yr`	removal	−0.1103	0.067	45/48	useful
`mnth`	removal	−0.0055	0.070	149/195	useful
`hr`	removal	−0.5684	0.491	443/537	useful
`holiday`	removal	−0.0000	−0.000	18/24	uninformative
`weekday`	removal	−0.0052	−0.001	100/121	useful
`workingday`	removal	−0.0162	0.000	63/71	useful
`weathersit`	removal	−0.0175	0.016	9/10	useful
`temp`	removal	−0.0143	0.165	42/49	useful
`atemp`	add back	0.0003	0.162	–	redundant
`hum`	removal	−0.0006	0.097	2/2	redundant
`windspeed`	removal	−0.0006	0.015	1/1	redundant

validation set, so no k-fold cross-validation was used. Ridge was selected by the highest validation R^2; Lasso and Elastic Net by the largest λ within 0.002 of the path's best R^2.

Results. In the first run all L1-type paths peaked at their last grid value; the paths were extended to $\lambda = 10^{-5}$ at the same spacing, the first 30 values reproduced exactly, and only the Elastic Net choice changed. Table 2 and Figure 3 summarise the result. Ridge was best (validation R^2 0.926237 against 0.926228 unpenalised), Lasso retained 466 of 564 weights at 0.9248, and Elastic Net selected $\rho = 0.1$. Phase 3 had left almost no variance to remove, so the penalty could only add bias: the top-decile shortfall rose from 53.8 to 56.9 bikes under Lasso. Column verdicts (Table 3) combine a grouped ablation, in which the design is rebuilt without the column and every term derived from it and refitted with the Ridge setting, or an add-back test; the R^2 of the column alone in its full encoding; and Lasso survival. A column is *useful* if its removal costs at least 0.002 in R^2, otherwise *redundant* if its own R^2 is at least 0.01, and otherwise *uninformative*. *holiday* retains most of its terms under Lasso yet costs nothing when removed: a surviving coefficient does not establish usefulness. **Passed on:** the 466 Lasso survivors (from 11 original columns) with lineage; Ridge as the recommended regression model.

2.5 Phase 5: logistic regression

Method. An hour was labelled high demand if its count exceeded the 75th percentile of training hours in the same hour × working-day × year group (96 cutoffs from the training portion, applied unchanged to validation). A single global cutoff would label 8 of the 24 hours as never high, so the classifier would mainly learn the clock; a cutoff without year would label 5.6% of 2011 and 43.1% of 2012 hours as high. With the chosen label the

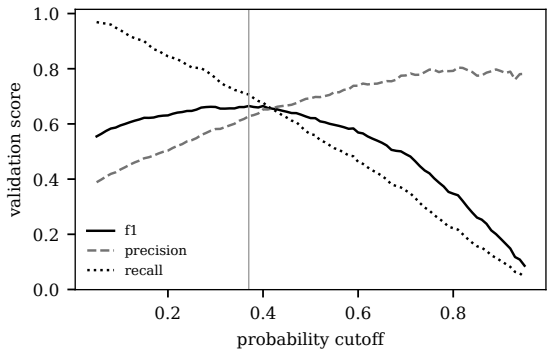


Figure 4: Phase 5 validation precision, recall and F-measure against the cutoff; vertical line: tuned cutoff.

Table 4: Phase 5 confusion matrices and scores (validation).

Cutoff	TP	FP	FN	TN	Acc.	Prec.	Rec.	F1
0.37	388	230	161	1,399	0.820	0.628	0.707	0.665
0.50	309	137	240	1,492	0.827	0.693	0.563	0.621

rates are 24.3% and 24.6%, and 2,127 training hours (24.4%) are positive (25.2% in validation). The inputs were exactly the 466 Lasso survivors with the fitted scaler. The log-loss $-\frac{1}{m}\sum[y\log p + (1-y)\log(1-p)]$ with $p = \sigma(Xw)$ was minimised by gradient descent using the explicit gradient $X^\top(p-y)/m$. Its curvature is at most $L = \lambda_{max}(X^\top X/m)/4 = 1.0424$, since $p(1-p) \le 1/4$, so the step $1.9/L = 1.823$ is safe. The probability cutoff maximised validation F-measure over 0.05–0.95.

Results. The fit converged in 1,229 updates. ROC-AUC was 0.871; at the tuned cutoff 0.37, accuracy was 0.820 against 0.748 for always predicting "not high", precision 0.628, recall 0.707 and F-measure 0.665 (Table 4, Figure 4). At 0.5, accuracy was higher (0.827) but the F-measure lower (0.621): 79 more busy hours were missed in exchange for 93 fewer false alarms. With a quarter of the hours positive, accuracy therefore rewards missing busy hours. Temperature had the largest odds ratio (×3.65 per standard deviation), followed by month indicators. The label still carries seasonality: 0.4% of January and 48.8% of June training hours are positive. A calendar-only ranking (month × weekday) reaches ROC-AUC 0.809, whereas the classifier reaches 0.747 within a single month (0.641–0.862 within a season), which measures its skill beyond the calendar.

3 Bonus: asymmetric cost

Under-prediction was assumed to cost three times as much as over-prediction of the same size, and the squared error was retained: $\ell(r) = \frac{1}{2}c(r)\,r^2$ with $r = \hat{y} - y$ and $c = 3$ for $r < 0$, otherwise $c = 1$. The derivative $c(r)\,r$ is continuous, so the gradient is $X^\top(c \odot r)/n = X^\top r/n + 2X^\top \min(r, 0)/n$, and its Lipschitz constant is $3\lambda_{max}$; the minimiser is the 0.75 expectile rather than the mean. A cost linear in the number of bikes would target the 75th percentile instead, but it has no gradient at zero, so fixed-step gradient descent could not converge on it. On Phase 1's features the Phase 1 learning rate diverged within 12 updates, whereas the rate divided by three (0.318) converged in 388 updates. Predictions rose by 42.6 bikes per hour on average. R^2 fell from 0.682 to 0.606 and RMSE rose from 103.6 to 115.3, as both measure the squared-error objective, while the asymmetric cost fell from 11,145 to 9,343 (16.2%). Bikes short per hour fell from 37.2 to 21.2, idle bikes rose from 39.6 to 66.1, and the top-decile shortfall fell from 176.6 to 112.1. Because Phase 1 underfits, much of the lift went to hours that were already over-predicted. On Phase 2's features the same loss shifted predictions by only 18.9 bikes, cost 0.016 in R^2 (from 0.923) and reduced the top-decile shortfall from 58.9 to 34.1 bikes.

4 Limitations

Seasonal label leakage. The Phase 5 label removes the level of each hour, day type and year but not the season; a large part of the classifier's ROC-AUC (0.871) is reproducible from the calendar alone (0.809). Adding season to the grouping would have produced many groups with too few rows for a stable percentile. **Growth extrapolation.** On the chronological split the models under-predict their own busiest hours (mean error −36.6 bikes in the top predicted decile for the target design, against −12.4 on the seeded split), since growth beyond the training period cannot be extrapolated. **One validation set.** λ, the Phase 3 target and the Phase 5 cutoff were tuned and scored on the same seeded validation set, so those scores are slightly optimistic. **Phase 4 grid revision.** The L1-type λ paths were extended after the first official run because their optima lay at the grid edge; the revision is recorded in the notebook and changed only the Elastic Net choice.

5 Final submission

The submitted predictions come from Phase 4's Ridge model ($\lambda = 0.00045$) on the 564-feature design. With every choice frozen on the seeded split, the validation rows had served their purpose of selection, and the model was refitted on all 10,886 labelled hours by the same procedure, with encoders, scaler and centring means refitted on the combined rows (the same 92 exact combinations were removed). The final fit converged in 2,053 updates (training R^2 0.9317, RMSE 47.34). The 574 test hours were predicted in the order of the sample file, and 35 negative predictions were set to zero, since rentals cannot be negative (mean prediction 191.8, maximum 870.9 bikes).